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Near Real-Time Crop Yield Forecasting with Interpretable Time-Series Deep Learning

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Why is crop yield forecasting important?

- If we know **in advance** that a crop failure is likely, we can organize aid efforts to prevent food insecurity *(You et al., 2017)*
- At a national level - yield predictions can inform import/export decisions and supply chains *(Khaki et al., 2020)*
- At a local level - farmers can utilize yield predictions to make better management decisions *(Horie et al., 1992)*

Climate change effects on crop yield

- Crop production is extremely sensitive to fluctuations in climate *(Ortiz-Bobea et al., 2018)*
  - Climate change has already reduced agricultural productivity growth by 21% *(Ortiz-Bobea et al., 2021)*
  - Critical to understand how crops will react

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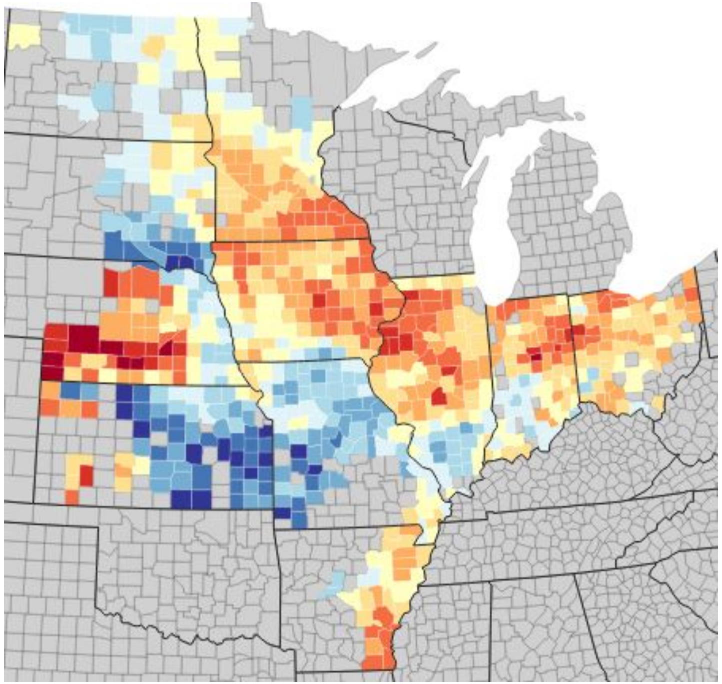
Existing approaches: field surveys

- Crop yield forecasts in the US often depend on **field surveys** *(Basso & Liu, 2019)*
  - Experts go into the field and observe crop status
  - Expensive, can’t be done frequently

Deep learning could help

- Crop yield is a complex function of weather, soil conditions, moisture, technology, and many other factors
  - In our dataset, we have 6000+ input features (covariates)
  - Hard to know which variables and time periods are relevant
- Ideally, deep learning methods can **learn** these relationships from data



Example ground-truth yield map from 2012. Figure from http://sustain.stanford.edu/crop-yield-analysis

Problem setting

Forecast county level yields for 6 crops, for **all US counties**

- Corn, upland cotton, sorghum, soybeans, spring wheat, winter wheat
- USDA provides ground-truth labels

Years: 1981-2020

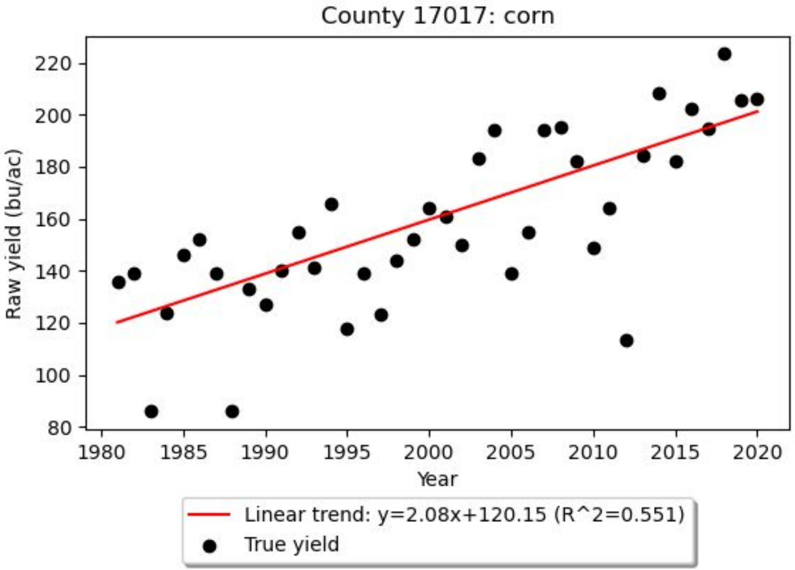
8-fold cross-validation (for each fold: hold out 5 years for validation & 5 years for testing)

Features for each county

- PRISM weather data **(daily)**
  - Temperature (min/max/mean), precipitation, humidity, etc
- NLDAS land surface model variables **(hourly)**
  - Soil moisture, moisture availability, soil temperature, etc.
- gSSURGO soil survey data **(for each soil layer; constant over time)**
  - Available water capacity, bulk density and electrical conductivity, pH, organic matter, soil texture type (silt/clay/sand), etc.
- USDA crop progress & condition reports **(weekly)**, irrigation **(constant)**
- Preprocessing
  - Time-varying features (PRISM, NLDAS) converted to **weekly**
  - All features aggregated to county-level (originally gridded)

Linear trend

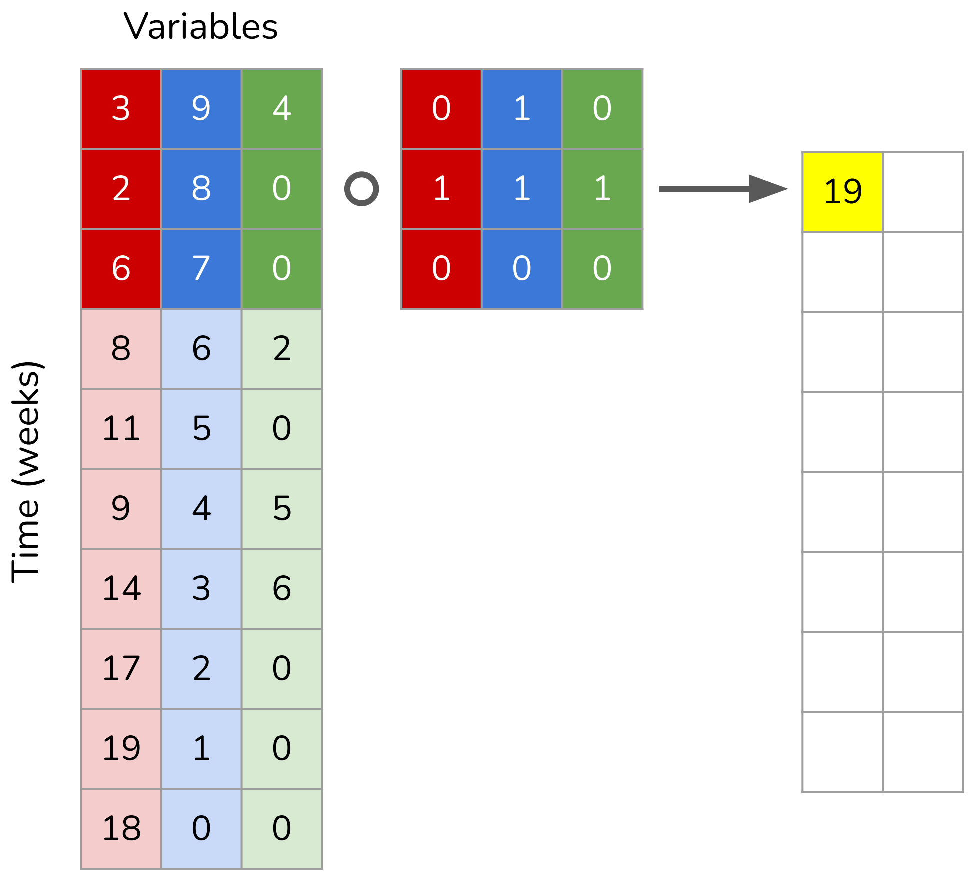
- Yields have generally been increasing due to technological improvements
- Fit a linear trend-line **for each county**
- The trend-line is used as a feature to predict the yield
- Evaluate model on how well it predicts *deviation from trend*



Key building block: 1D CNN

- Weather data is **sequential**: for each variable, there is a value **for each week**
- Deep learning architectures for sequential data
  - Recurrent neural networks (RNN), including LSTM/GRU
  - Transformer architecture
  - **1D Convolutional neural networks (CNN) - over time**
- So far, **1D CNNs** seem to work the best
  - Used in *(Khaki et al. 2020)*
  - Computes features on small time-windows first, then hierarchically aggregates them together. Each time-window has a contribution
  - Implemented in PyTorch

How 1D CNN works



Slide each filter over the input time-series.

At each position, take the "dot product" of the input & the filter.

Filter weights are learned from data!

How 1D CNN works

Variables

Time (weeks)

3	9	4
2	8	0
6	7	0
8	6	2
11	5	0
9	4	5
14	3	6
17	2	0
19	1	0
18	0	0

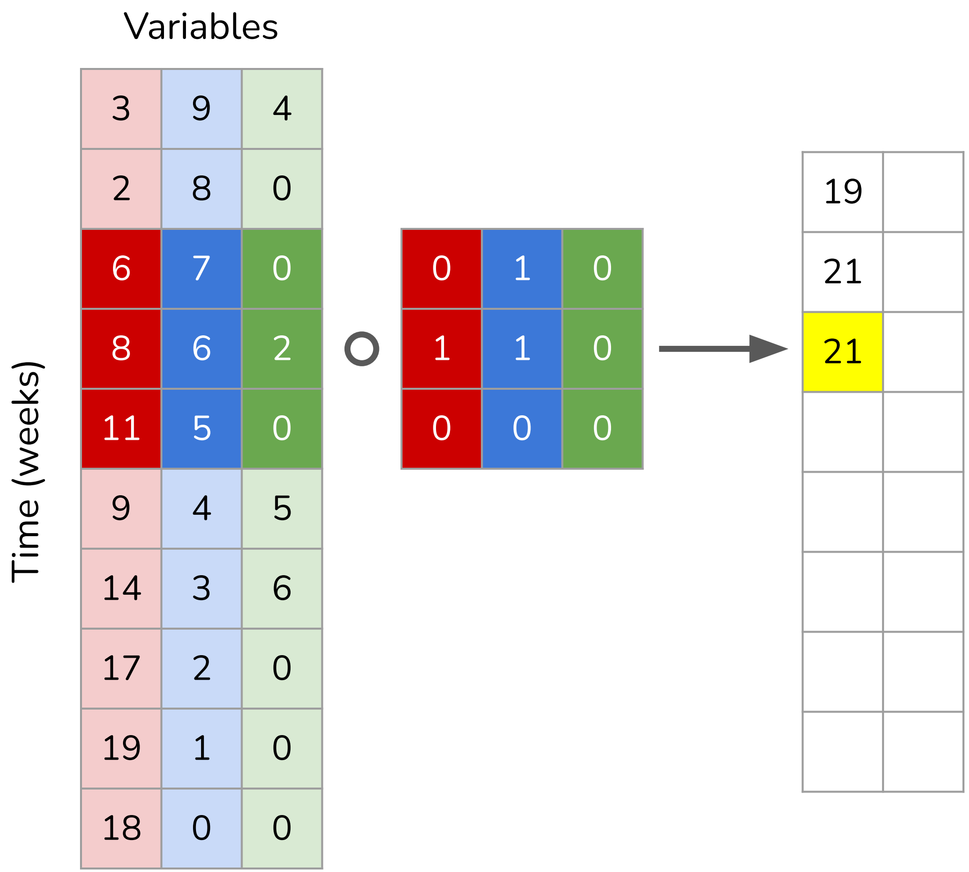
$\circ$

0	1	0
1	1	0
0	0	0

$\rightarrow$

19	
21	

How 1D CNN works



How 1D CNN works

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∘

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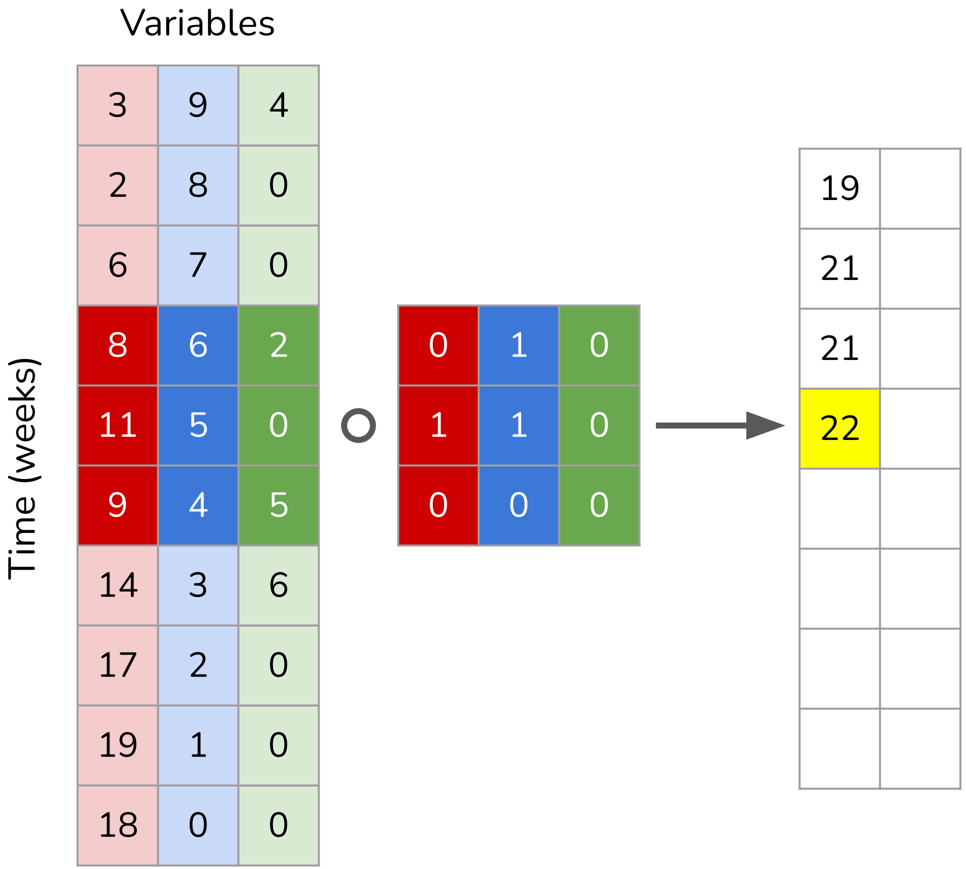
0	0	0

→

19	
21	
21	
22	

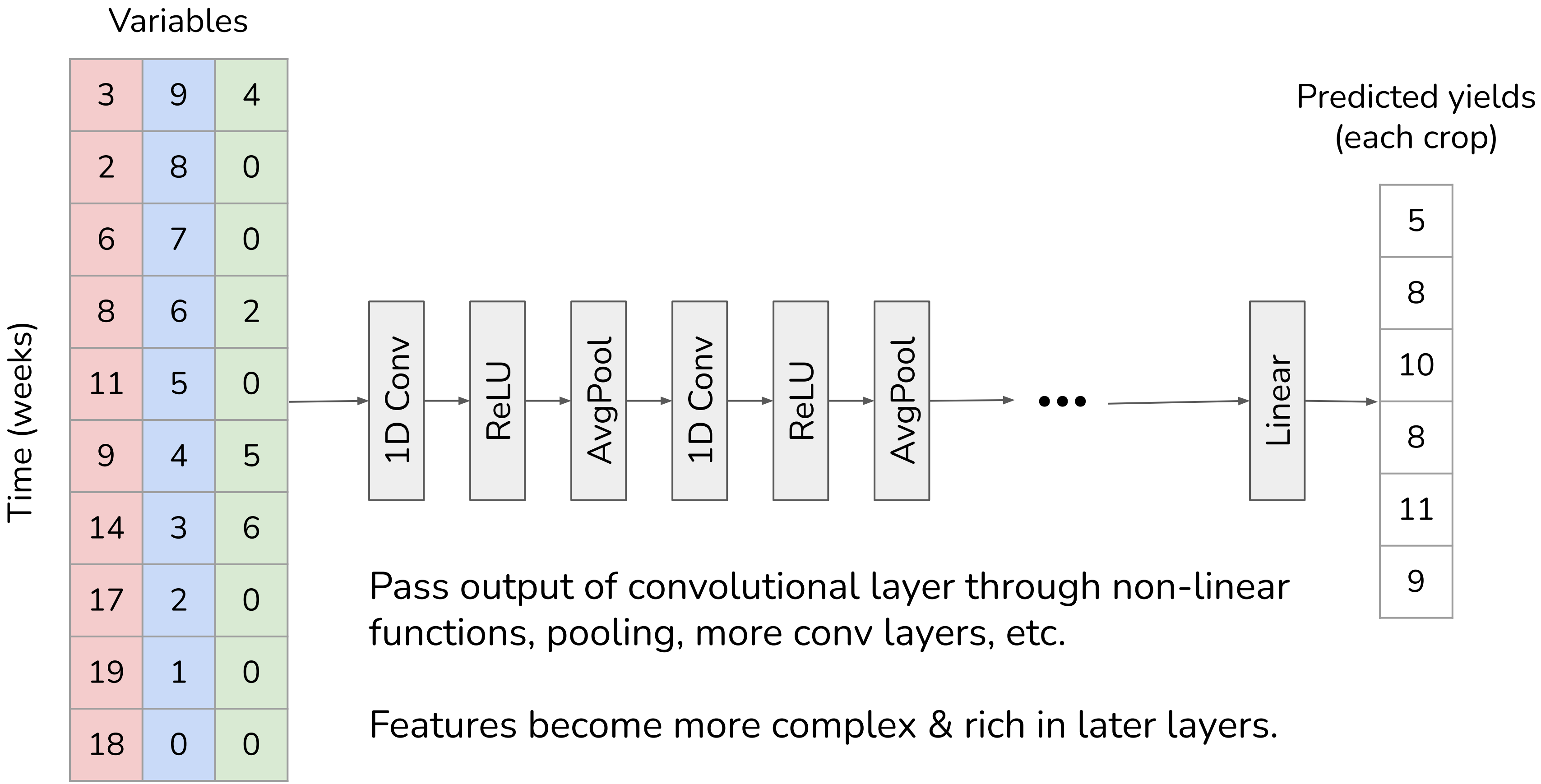
How 1D CNN works



Keep applying this filter down the sequence.

Then go to the next filter (column)

How 1D CNN works



Pass output of convolutional layer through non-linear functions, pooling, more conv layers, etc.

Features become more complex & rich in later layers.

Learning (parameter estimation)

As in Linear Regression, our goal is to find model parameters ("coefficients") that minimize sum of squared **errors** (between predicted & true yield)

Variables

Time (weeks)

3	9	4
2	8	0
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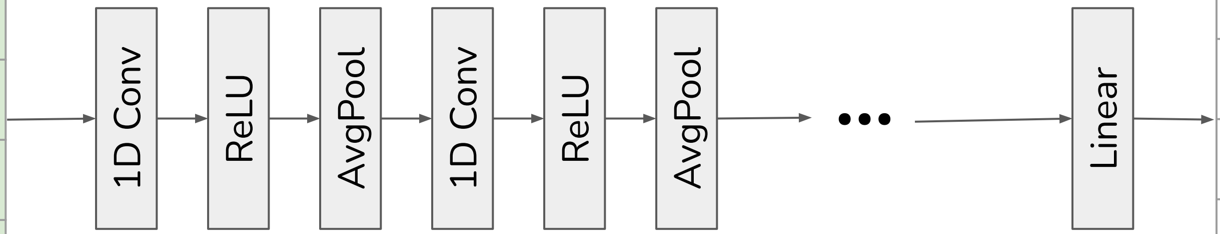

Predicted yields (each crop)

5
8
10
8
11
9

Learning (parameter estimation)

As in Linear Regression, our goal is to find model parameters (“coefficients”) that minimize sum of squared **errors** (between prediction & true yield)

Variables

Time (weeks)

3	9	4
2	8	0
6	7	0
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11	5	0
9	4	5
14	3	6
17	2	0
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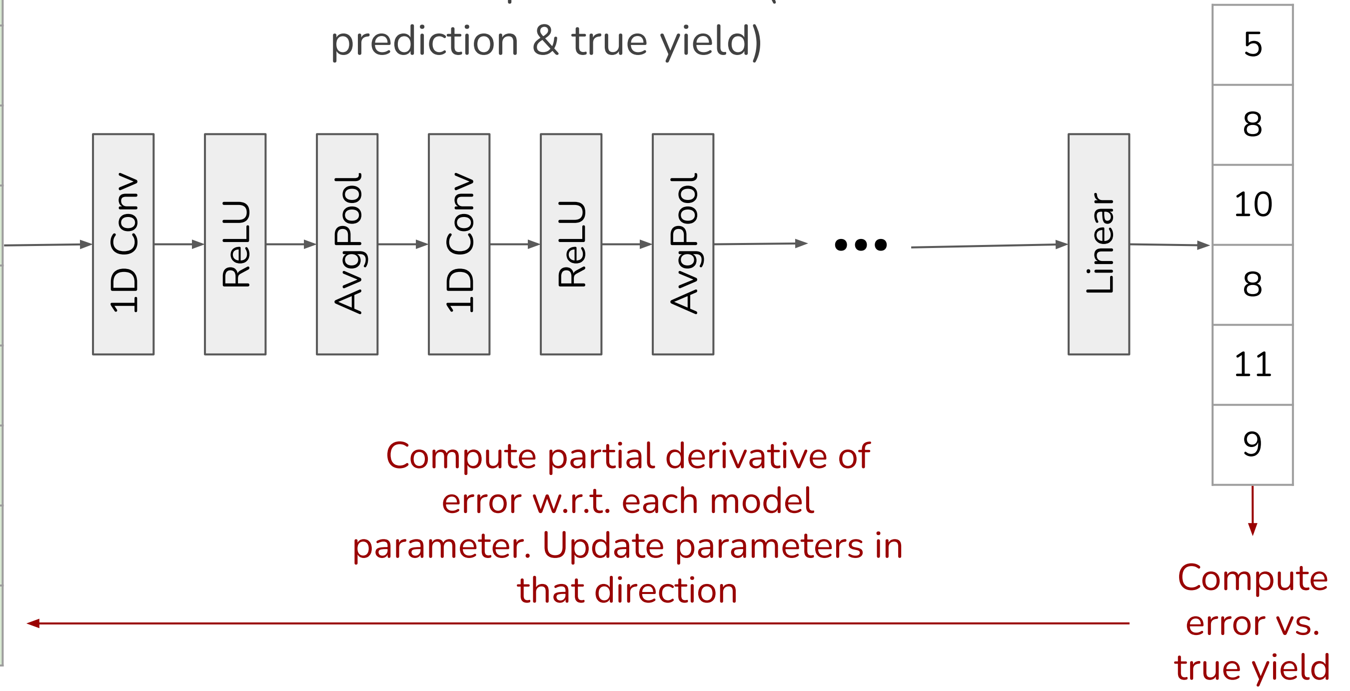
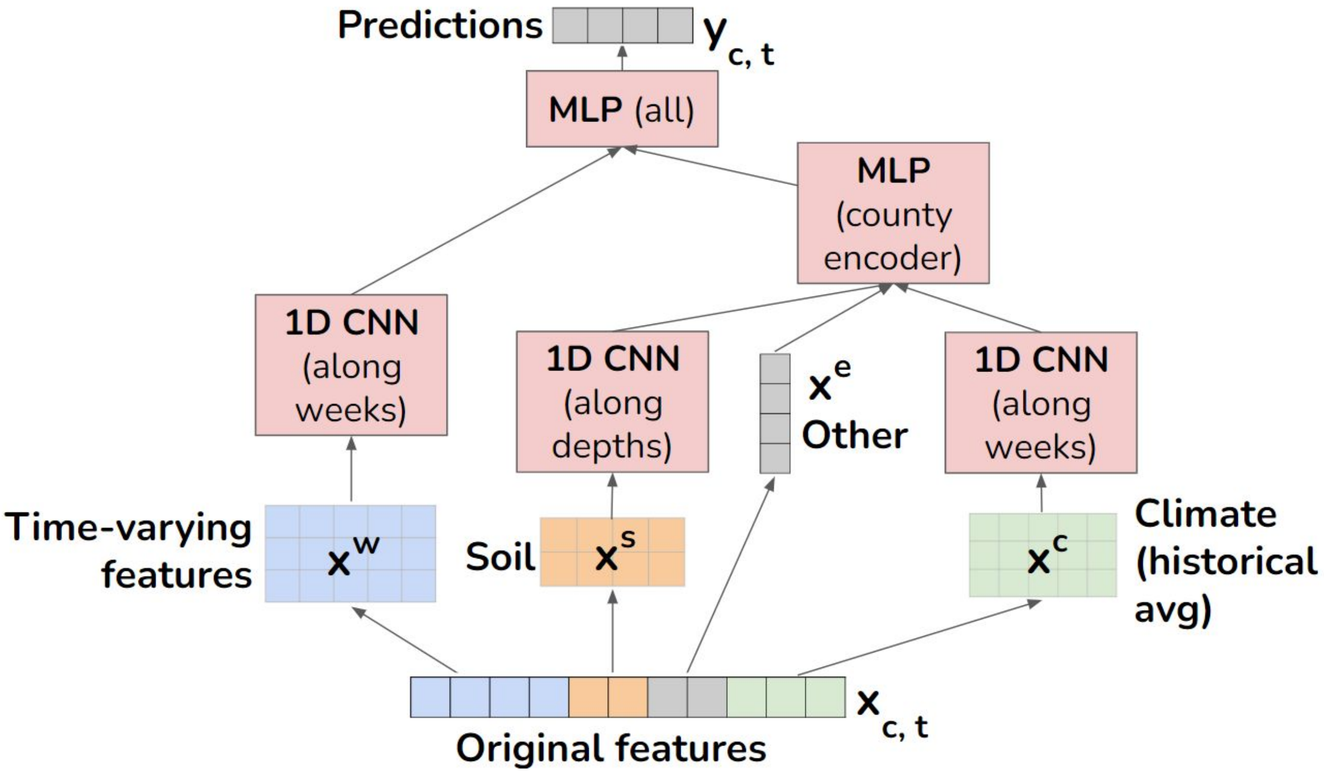
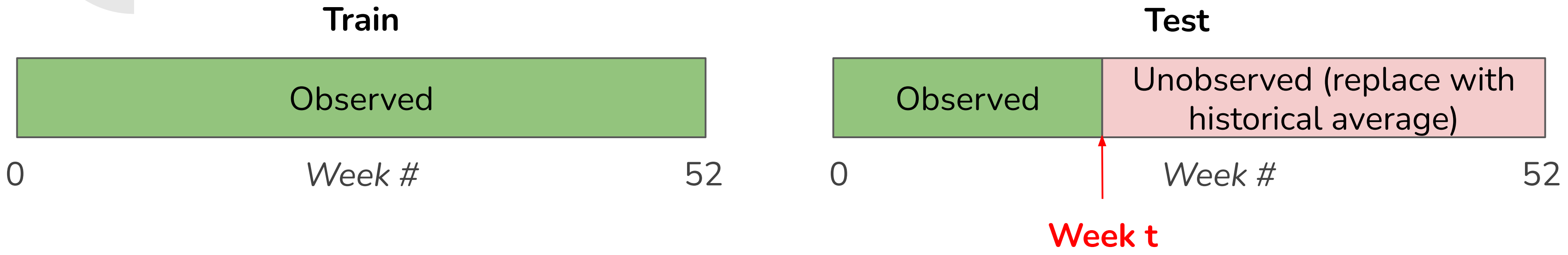
Predicted yields (each crop)



Overall model structure



How to make forecasts mid-year



- Of course this is sub-optimal: it doesn't make sense for weather conditions to suddenly revert to average at week *t*.
- Future work: use **weather forecasts** to predict future conditions

Comparison of methods: full-year prediction (all years, out-of-sample cross-validation R^2)

Method	Corn, R^2	Soybeans, R^2
CNN	**0.479**	**0.474**
LSTM	0.454	0.448
GRU	0.392	0.408
Transformer	0.401	0.383
MLP (all features, no temporal structure)	0.322	0.351
Ridge regression (monthly GDD/KDD/soil moisture, fitted per state)	0.192	0.164

Aside: graph neural networks for spatial correlations

In previous work, we used a graph neural network (GNN) method, GraphSAGE *(Hamilton et al, 2018)*, to allow each county's feature representation to be informed by **spatial context** (aggregating features from nearby counties)

- Each county is a node, and edges connect neighboring counties

However in our current setting, GNN does not help. Now that we are using per-county trend, county avg climate, and USDA crop progress reports, these features are enough to ensure that neighboring counties' predictions are correlated, without having to explicitly use graph structure. More investigation is needed!

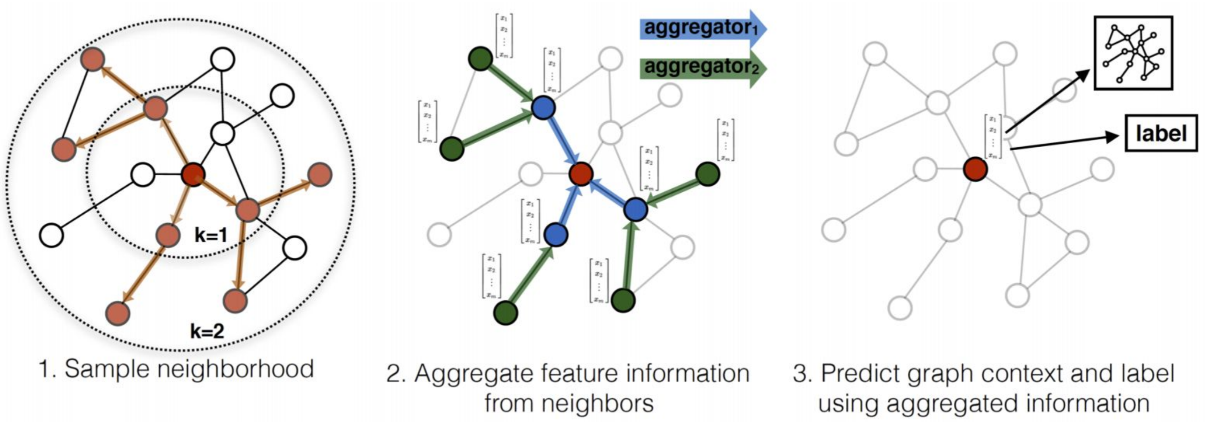
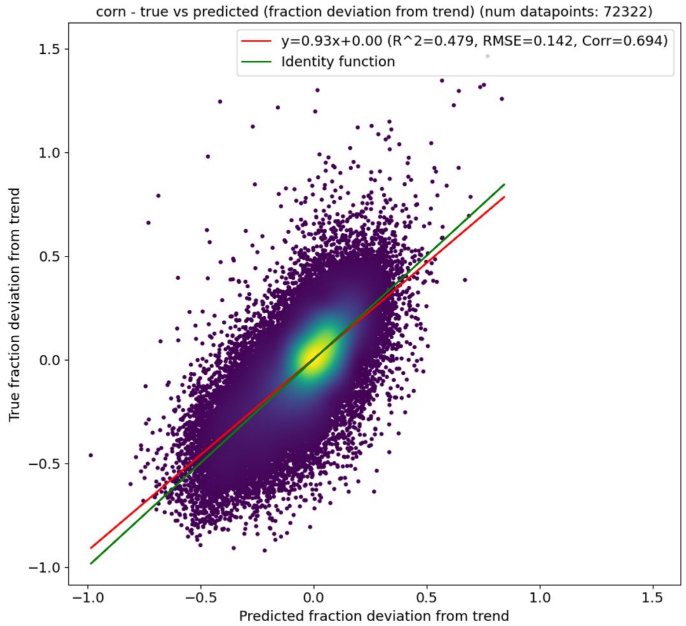
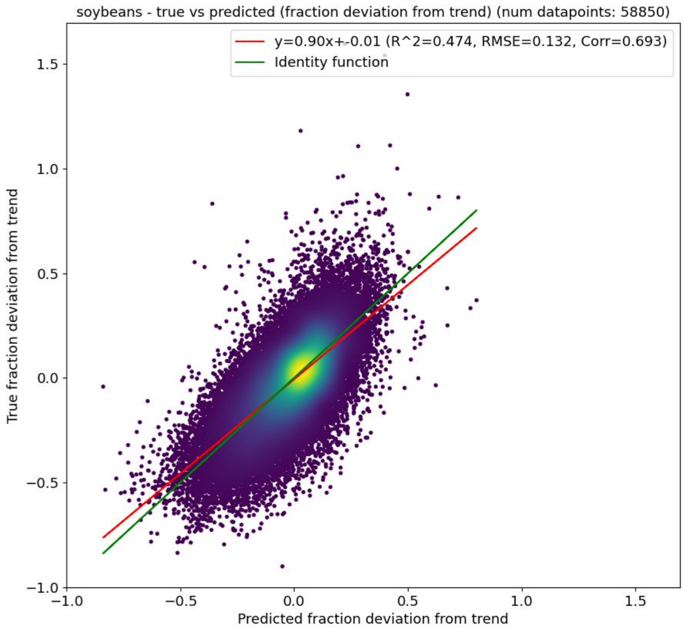


Figure from Will Hamilton et al. Inductive Representation Learning on Large Graphs. *Advances in Neural Information Processing Systems*, pages 1024–1034, 2017.

True vs predicted (<u>all years & counties</u>), end of year / out-of-sample

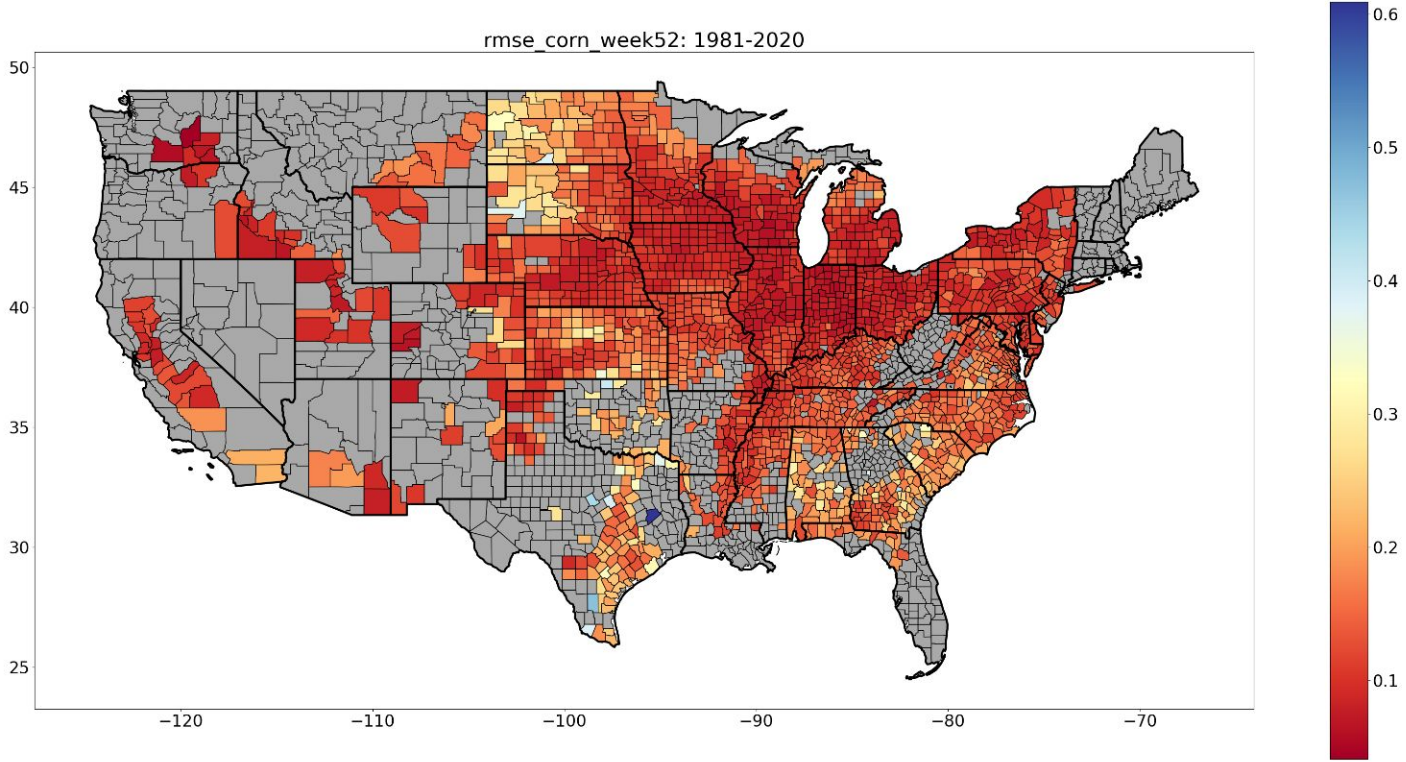
Note: we are evaluating with **fraction deviation** from trend



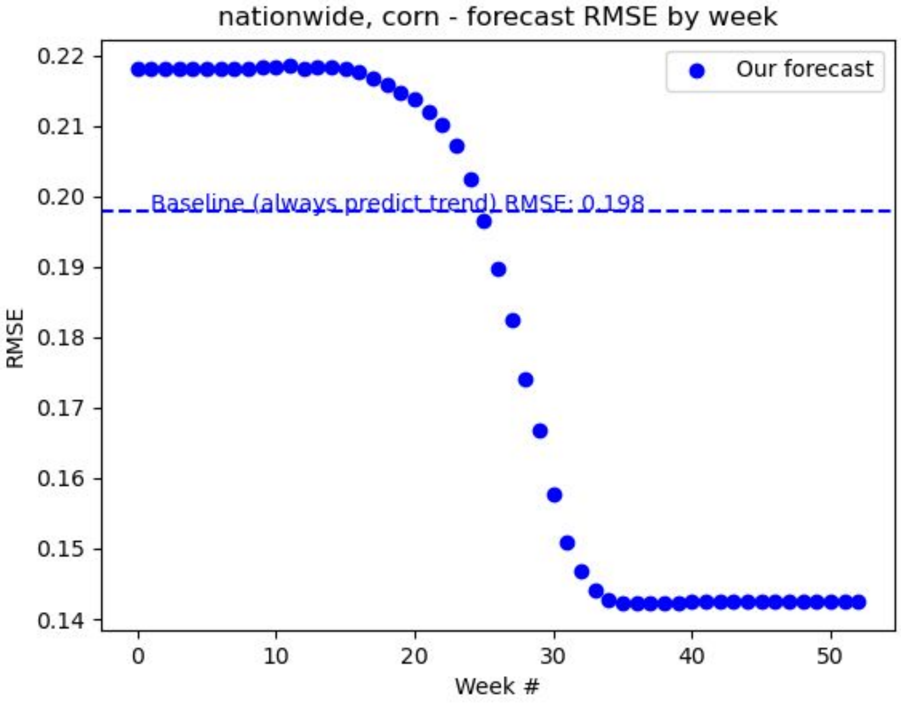


RMSE by county (corn, all years, end of year)

Model accuracy is heterogeneous by county; does better in Corn Belt



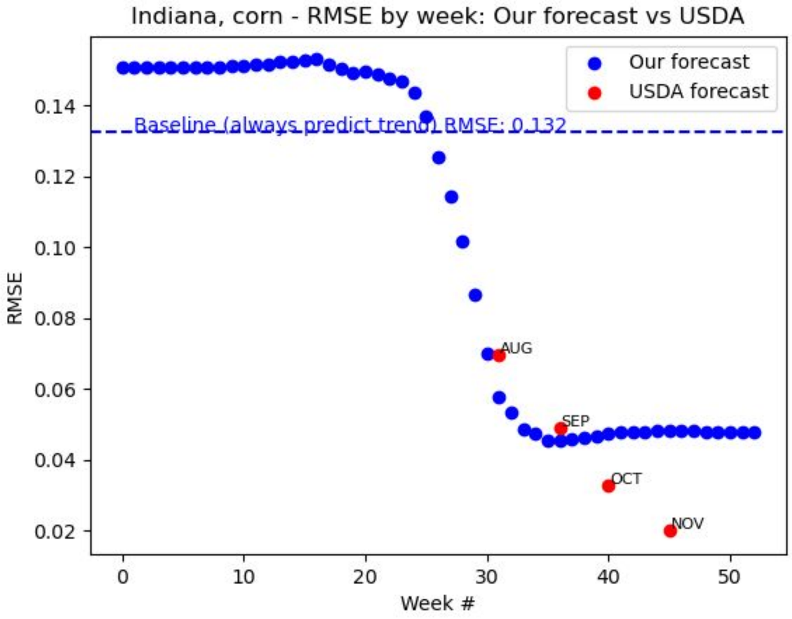
RMSE by week (corn, all years)



Comparison with USDA forecast

When aggregated to state level, our forecasts are roughly comparable to USDA's forecasts in key Corn Belt states for August/September

- **Very preliminary result, not peer-reviewed yet**

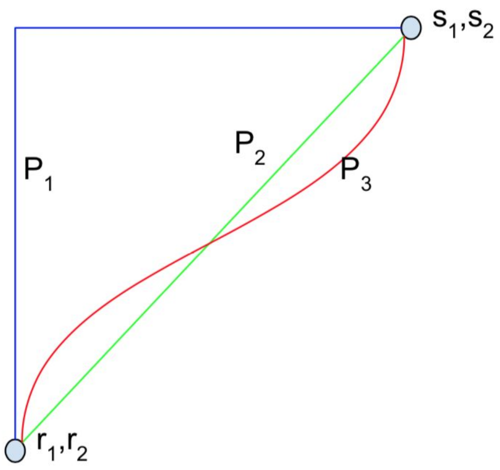


Interpreting model behavior

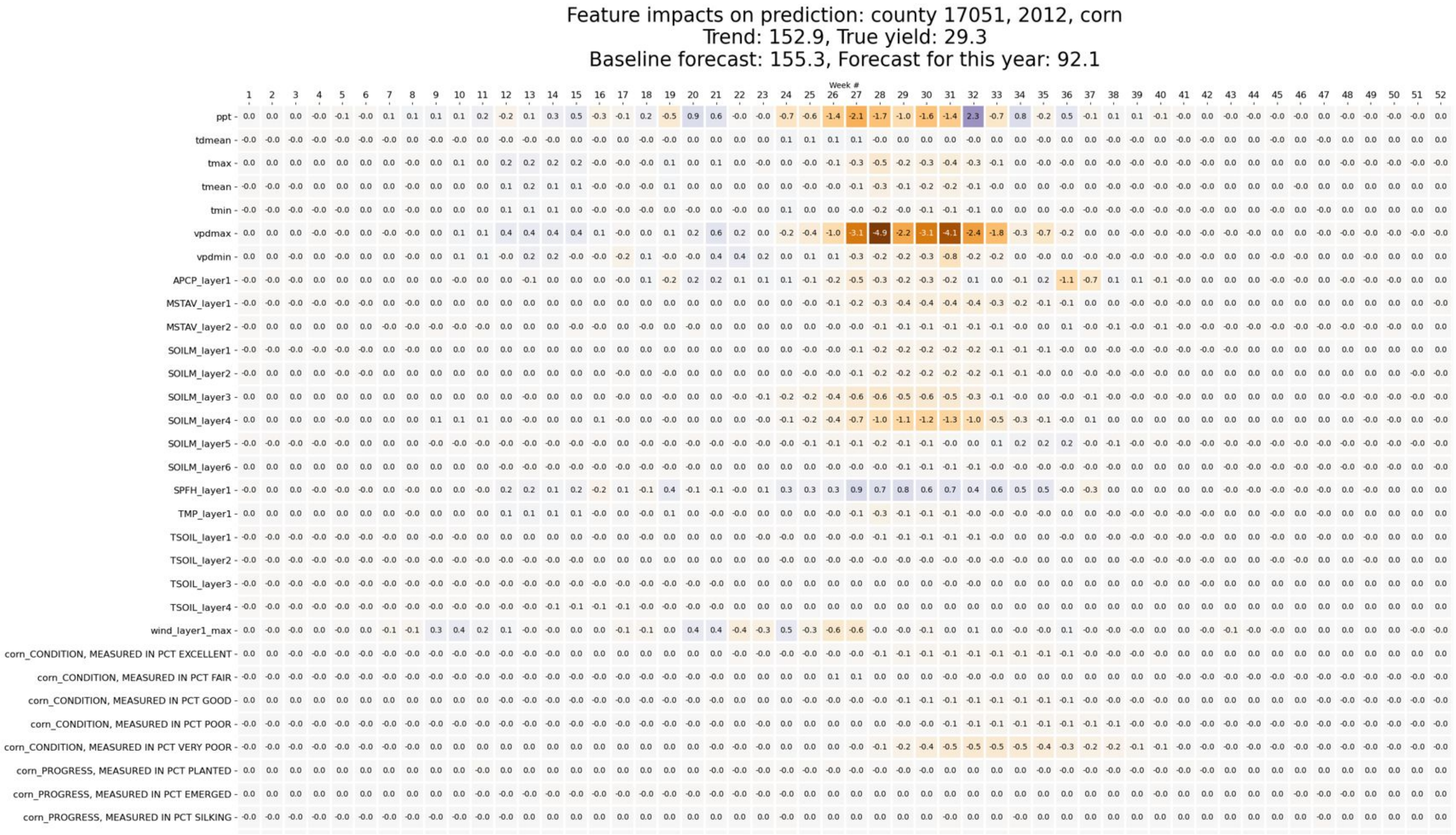
- Calculate contribution of each **week** by seeing how much forecast changes between week *t* and *t+1*
- But which **input variables** caused the forecast to shift most? We use the method **Integrated Gradients** *(Sundararajan et al. 2017)*
  - Explains forecast of a datapoint compared to a neutral baseline. How much did each feature cause the forecast to shift from the baseline?
  - **Baseline:** given county/year, with variables from week *t* replaced with county's historical average
  - **Input:** given county/year, with variables from week *t+1* replaced with county's historical average

How Integrated Gradients works

- Draw a line: (**baseline** → **input**)
- At each point on the line, take the gradient of the **forecast** with respect to each **input variable.**
    - Tells you how changing the input variable would impact the forecast
    - Integrate the gradient along this line
- Variable attributions **sum to the total difference between forecast(baseline)** → **forecast(input)**

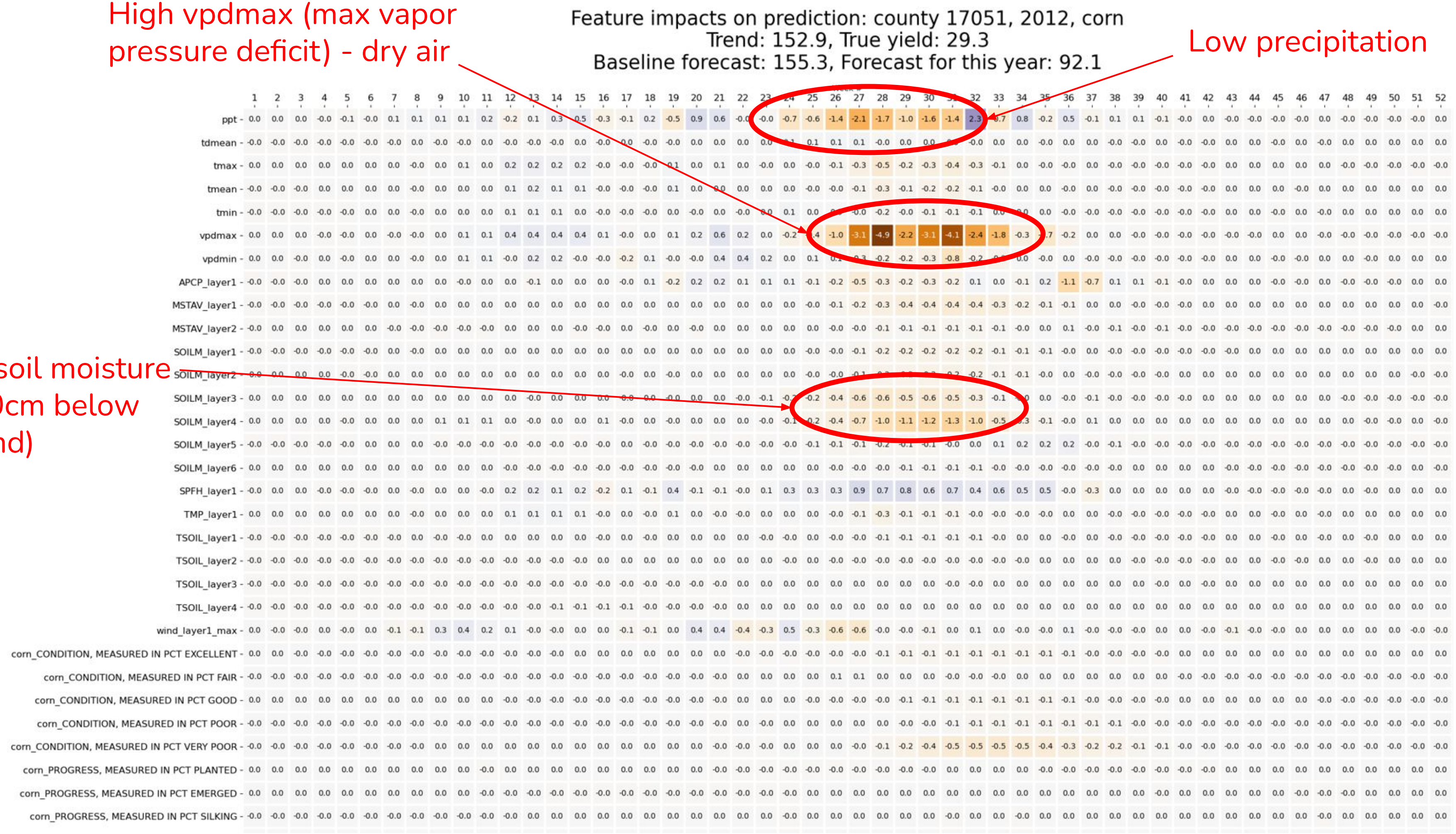


$$\text{IntegratedGrads}_i(x) ::= (x_i - x'_i) \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} \, d\alpha$$


Feature impacts on prediction: county 17051, 2012, corn
Trend: 152.9, True yield: 29.3
Baseline forecast: 155.3, Forecast for this year: 92.1
Week #
ppt
tdmean
tmax
tmean
tmin
vpdmax
vpdmin
APCP_layer1
MSTAV_layer1
MSTAV_layer2
SOILM_layer1
SOILM_layer2
SOILM_layer3
SOILM_layer4
SOILM_layer5
SOILM_layer6
SPFH_layer1
TMP_layer1
TSOIL_layer1
TSOIL_layer2
TSOIL_layer3
TSOIL_layer4
wind_layer1_max
corn_CONDITION, MEASURED IN PCT EXCELLENT
corn_CONDITION, MEASURED IN PCT FAIR
corn_CONDITION, MEASURED IN PCT GOOD
corn_CONDITION, MEASURED IN PCT POOR
corn_CONDITION, MEASURED IN PCT VERY POOR
corn_PROGRESS, MEASURED IN PCT PLANTED
corn_PROGRESS, MEASURED IN PCT EMERGED
corn_PROGRESS, MEASURED IN PCT SILKING

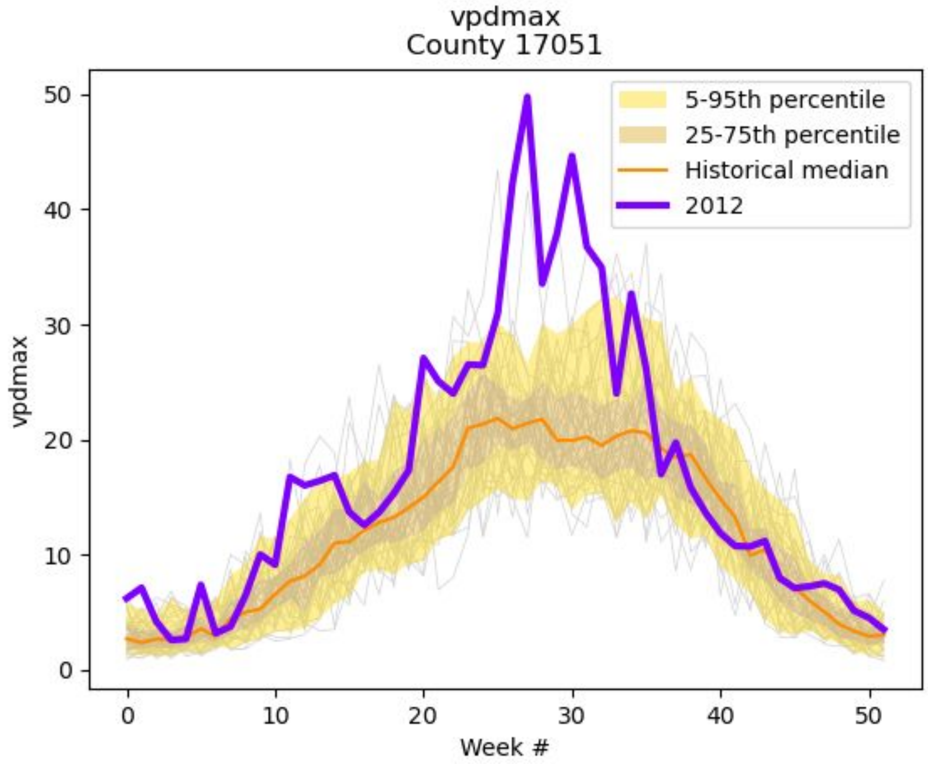
High vpdmax (max vapor pressure deficit) - dry air

Low precipitation

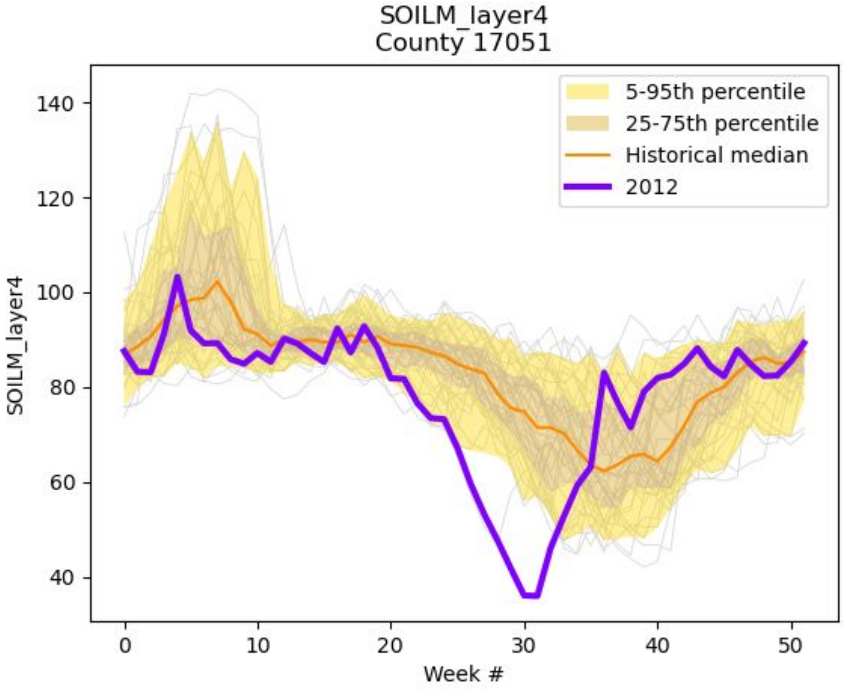


Low soil moisture (0-40cm below ground)

Vpdmax was extremely high



Soil moisture was extremely low



Caveats

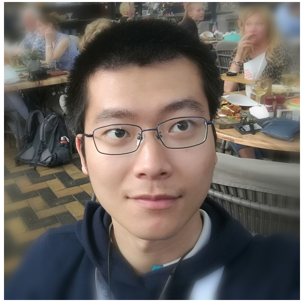
- Deep learning interpretability methods aren't always accurate (still an active area of research)
    - The method here doesn't capture how model uses **interactions** between variables
    - Only explains prediction on a single example, not globally
    - Hard to see if model is actually picking up concepts such as "growing degree days"
- Model accuracy is far from perfect - probably missing some things

Conclusion

- We developed deep learning techniques to forecast crop yields from a complex array of weather, soil, and management data
    - Convolutional networks seem to work best
- For corn and soybean, model can capture 45-50% of the variation in yield (across space and time) that’s not explainable due to technological advancements
- Model performs comparably to USDA forecast in Aug/Sep
- Using interpretability techniques helps us decipher what the model learned

Acknowledgements - Thank you!

Co-authors: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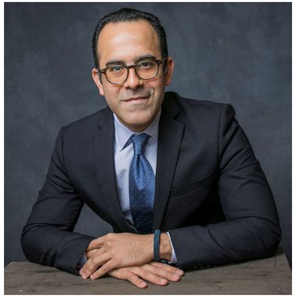
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